$\boxed{\text{03-12-2025}}$

$\rightarrow$ use case for Wald similar to F-test

$\rightarrow \theta = r(\beta) \Rightarrow \hat{\theta} = r(\hat{\beta})$

$\hookrightarrow \sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \underline{V}_\theta) \Rightarrow \underline{V}_\theta = ?$

$\rightarrow$ Ex: Regression Intervals:

$m(X) = E[Y|X] = X^T\beta$

$m(x) = x^T\beta \Rightarrow \hat{m}(x) = x^T\hat{\beta}$

$\sqrt{n}(\hat{m}(x) - m(x)) \xrightarrow{d} N(0, V_m)$

$\sqrt{n}(\hat{m}(x) - m(x)) = \sqrt{n}(x^T\hat{\beta} - x^T\beta) = x^T\sqrt{n}(\hat{\beta} - \beta)$

$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \underline{V}_\beta) \Rightarrow x^T\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, x^T\underline{V}_\beta x)$

$\hat{V}_M = x^T\hat{\underline{V}}_\beta x \Rightarrow \hat{\underline{V}}_\beta = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right)^{-1} \hat{\underline{\Omega}} \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right)^{-1}$

$\hat{C} = [x^T\hat{\beta} \pm 1.96(\text{s.e. } \hat{m}(x))]$

s.t.

$se(\hat{m}(x)) = \frac{\sqrt{\hat{V}_m}}{\sqrt{n}}$

→ Case 2: r is non-linear

Delta Method: (need r is cts. diff in a nbhd of β)

$$\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0, \underline{V}_\theta)$$

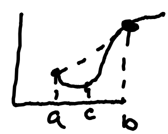
s.t.

$$\underline{V}_\theta = \underline{R}^T \underline{V}_\beta \underline{R}$$

$$\underset{k\times q}{\underline{R}} = \nabla r(\beta) = \left.\frac{\partial r(b)^T}{\partial b}\right|_{b=\beta}$$

Mean Value Thm: if f is a cts function on $[a,b]$ and diff'able on (a,b), $\exists\, c \in (a,b)$ s.t.

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$



$$\Rightarrow f(b) = f(a) + f'(c)(b-a)$$

$$r(\hat{\beta}) = r(\beta) + \nabla r(\bar{\beta})^T(\hat{\beta}-\beta)$$

s.t.

$\bar{\beta}$ is a vector "b/w" $\hat{\beta}, \beta$

$$\Rightarrow \sqrt{n}(\hat{\theta}-\theta) = \nabla r(\bar{\beta})^T \sqrt{n}(\hat{\beta}-\beta)$$

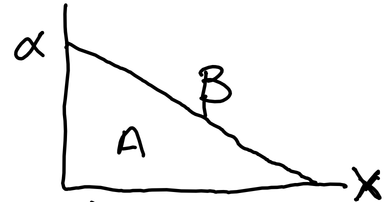
$$= \nabla r(\beta)^T \sqrt{n}(\hat{\beta}-\beta) + \underbrace{\underbrace{(\nabla r(\overset{\text{converge to } \beta}{\bar{\beta}}) - \nabla r(\beta))^T}_{\text{by CMT } o_p(1)} \underbrace{\sqrt{n}(\hat{\beta}-\beta)}_{O_p(1)}}_{o_p(1)}$$

$$= \nabla r(\beta)^T \sqrt{n}(\hat{\beta}-\beta) + o_p(1) \xrightarrow{d} N(0, \nabla r(\beta)^T \underline{V}_\beta \nabla r(\beta))$$

Ex: Consumer Surplus:

$$Y = \alpha + \beta X + e$$

s.t. $\alpha > 0,\ \beta < 0$



want. $\hat{A}, \hat{\alpha}, \hat{\beta}$

$$\hat{A} = -\frac{\hat{\alpha}^2}{2\hat{\beta}} \quad \text{(DELTA)}$$

$$\sqrt{n}\,(\hat{A} - A) \xrightarrow{d} N(0, V)$$

$$A = r(\alpha, \beta) = -\frac{\alpha^2}{2\beta}$$

MVT:

$$\Rightarrow \hat{A} = r(\hat{\alpha}, \hat{\beta}) = r(\alpha, \beta) + \nabla r(\bar{\alpha}, \bar{\beta})^T \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix}$$

$$\nabla r = \begin{bmatrix} \frac{\partial r(a,b)}{\partial a} \\ \frac{\partial r(a,b)}{\partial b} \end{bmatrix}\Bigg|_{\substack{a=\bar{\alpha},\\ b=\bar{\beta}}} = \begin{bmatrix} -\frac{a}{b} \\ \frac{a^2}{2b^2} \end{bmatrix}\Bigg|_{\substack{a=\bar{\alpha}\\ b=\bar{\beta}}}$$

$$\sqrt{n}\,(\hat{A} - A) = \begin{bmatrix} -\frac{\bar{\alpha}}{\bar{\beta}} \\ \frac{\bar{\alpha}^2}{2\beta^2} \end{bmatrix}^T \sqrt{n} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix}$$

$$= \begin{bmatrix} -\frac{\bar{\alpha}}{\bar{\beta}} \\ \frac{\bar{\alpha}^2}{2\bar{\beta}^2} \end{bmatrix}^T \sqrt{n} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix} + \left(\begin{bmatrix} -\frac{\bar{\alpha}}{\bar{\beta}} \\ \frac{\bar{\alpha}^2}{2\bar{\beta}^2} \end{bmatrix} - \begin{bmatrix} -\frac{\bar{\alpha}}{\bar{\beta}} \\ \frac{\bar{\alpha}^2}{2\bar{\beta}^2} \end{bmatrix} \right) \sqrt{n} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix}$$

$$\xrightarrow{d} N(0, V) : V = \begin{bmatrix} -\frac{\alpha}{\beta} \\ \frac{\alpha^2}{2\beta^2} \end{bmatrix} ; \quad V_{\beta} = \begin{bmatrix} -\frac{\alpha}{\beta} \\ \frac{\alpha^2}{2\beta^2} \end{bmatrix}$$

$$\hat{V} \xrightarrow{p} V$$

→ Bootstrap:

↳ algorithm: non-parametric bootstrap

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① Construct Bootstrap Sample by making n i.i.d draws w/ replacement from original sample. (y_i^*, x_i^*)

$\{(y_i^*, x_i^*)\}_{i=1}^{n}$:= bootstrap sample

② Construct bootstrap estimate $\hat{\theta}^*$ by applying w/e approach used to est. $\hat{\theta}$ to the bootstrap sample from step 1.

↳ reg $\hat{\theta}$ in new sample $\Rightarrow \hat{\theta}^*$

B - # of BS samples

$\hat{\theta}_b^*$ for $b = 1, \ldots, B$